Warrant Containment in Dependency Graphs

What falls when a ground falls — and mostly negative results about when the answer can be trusted

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> **What changed from v1.0** (deposited 2026-08-17, 10.5281/zenodo.21981727). **One correction, in the abstract only.** v1.0 called the instrument a "three-valued calculus". That is a term of art meaning three TRUTH values, which places a system among K3, LP and Łukasiewicz's $Ł_3$ — and this one is not there. It is **two-valued**: every verdict is classical, and the third symbol is a mark on an *input*, barred from the value of any compound. §1 stated it correctly; the abstract did not.
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> **Nothing else changed.** No measurement, no table, no figure, no limit, and nothing in §3.6 or §6 — the sections a reader has cited. A citation to v1.0 remains accurate for everything except that one phrase.

Abstract

A conclusion rests on grounds. When a ground is withdrawn — a document is forged, a certificate lapses, a commander is lost — some conclusions stop standing and others do not, and which is which is currently something a human being remembers. This note reports a machine that computes it instead, and then reports, at greater length, the conditions under which its answer is worth having.

The instrument is small: a **three-symbol, two-valued** calculus whose core is machine-checked in Lean 4 on an empty axiom list — and a ledger over it in which every quantity carries the ground it rests on. From those, withdrawal propagates by arithmetic rather than by memory. (v1.0, deposited as 10.5281/zenodo.21981727, said "three-valued" here. That is a term of art meaning three TRUTH values, which would place this calculus among K3, LP and Ł3; it is not there. There are exactly two truth values and the third symbol is a mark on an input, barred from the value of any compound. §1 had it right and the abstract did not.)

The measurements are simulations, and the useful ones are negative. A cascade at a hundred thousand agents is cheap and, taken alone, useless: a randomly located loss takes down most of the collective. Redundancy does **nothing whatever** against a chosen loss — at ninety-five per cent redundancy an adversary aiming at the root still takes everything, and no density or redundancy tried brings that column below 0.791 (§3.9). Against a *random* loss redundancy works only at levels no real system has: anchored on the one real dependency graph measured here — its density, its declared redundancy — **a random loss costs 93% of the collective as well** (§3.9). Containment appears above roughly seventy-five per cent redundancy; real systems sit at the other end of that axis. Declared redundancy resting on a shared origin reports safety it does not provide. Permission and support are different requirements, so a collective with perfect evidence redundancy still dies of one commander. And an incomplete dependency map is wrong in a direction that depends on which edge is missing — a hidden shared origin flatters, a hidden alternative frightens — which is a correction to an earlier claim of one direction for both.

Twelve predictions made in the course of this work were refuted by its own runs — six by the author, six more by adversarial review. One of those refutations was **itself refuted** four rounds later (§3.4), which is reported with the rest. The corrections they forced are larger than the results they left standing. Every number below that is entered into `paper/prose-atoms.txt` is printed by a program in a public repository and re-checked by a regression suite of 121 stands.

1. The instrument

ZTL (Zero-Trust Logic) is a three-symbol calculus generated by a single principle: *truth is not granted on credit*. A connective returns T only if T is forced under every classical reading of an unverified mark; otherwise F. The mark itself never appears in a compound — it lives on atoms and evaporates at the first operator, so verdicts are two-valued while the *reason* for a verdict remains inspectable. The core is machine-checked in Lean 4 and prints an empty axiom list.

The **ledger** stores claims and grounds and never verdicts: a verdict is recomputed on every reading, because a stored one is a judgement taken on credit from a past moment. A quantity's ground may be an opaque external name (`inv-17`), another claim (`claim/c1`), a clocked certificate (`expiring/cert-9`), or an act with no inputs (`performed/x`). Grounds may be declared as alternatives (`inv-17|inv-18`), and since this work, a ground carries a **dimension**: whether it *supports* a claim or *permits* it.

Three operations follow without further rules.

Retraction travels. Withdraw a ground and the whole subtree moves, including claims that never named it.

Cost is a bracket, never a number. `cost(book, ground)` returns both readings — the ledger believed, and every declaration of independence assumed false — and the width between them. There is deliberately no function returning that cost as a bare figure: a number quoted without its width is a number quoted without the thing that makes it trustworthy.

Exposure is by unit. `exposure` reports *how much* rests on a ground rather than how many claims do, and never sums across incommensurable units. Adding metres to roubles is the error the numeric floor exists to refuse.

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2. What an ordinary store cannot say

One accounting scenario was loaded twice — plain SQLite, and the same facts with a warrant beside every value — and eight questions an auditor asks were put to both. Plain SQL **answers 2**, **answers 3 wrongly**, and **cannot express 3**. The middle category is the finding: a confident total over a mixture of evidenced and merely-stated figures, with nothing marking it.

Scored against the profession's own list rather than ours — the twelve ISA 315 assertions and eight accounts-payable procedures — **5 of 20** need a warrant at all. That count is not a measurement: the list is the profession's, the verdict beside each row was typed by the author, and the program counts his labels. The same applies to the 2/2/4 below. They are structured judgements made in public and open to correction, and are worth exactly what a reader thinks the labels are worth. Most of an auditor's day is ordinary querying. An earlier draft said completeness is answerable by **no ledger whatever**, because the invoice nobody entered leaves no row to carry a warrant. That is wrong, and the probe it was drawn from contradicts it: completeness for payables is not tested from the payables ledger but by reconciling that population against others — subsequent cash disbursements, unmatched goods-received notes, open purchase orders, vendor statement reconciliations, document-sequence checks, the three-way match. Each of those is a ledger, and each detects rows that should be there from rows that are. What is true and much narrower: a population cannot certify its own completeness from inside itself.

Against documented failures the instrument scores **2 TOUCHES, 2 EXPOSES, 4 OUT** of eight — again the author's labels, counted by a program — and *prevents none of them*. Wirecard's €1.9bn rested on forged bank letters; a forged ground is invisible here by construction, and the ledger would have printed a clean EARNED for years. Against system errors where nobody lied it does better, but less well than an earlier draft claimed: **1 caught, 1 caught only if declared, 1 not caught.**

The Mars Climate Orbiter is the one that shrank. Given both quantities with their units the floor refuses — `cannot compare 'lbf' with 'N'` — but that is not the case as it happened. The SM_FORCES file carried numbers; the unit lived in an interface specification. Run *that* and the floor returns **EARNED**: a quantity with no unit unifies with anything, because `_unify_units` compares two strings and a missing string matches. The instrument does not detect the absence of a unit. It demonstrates the value of carrying one, which is a design rule and not a catch, and whether values carry units is upstream of anything here.

The third case is not a documented one at all. It is a shape the author constructed — a conclusion, an input beneath it, a later correction — with figures echoing the high-debt-threshold episode. The retraction in it is stipulated rather than reconstructed, and the actual failure in that episode was a spreadsheet range selection, which is the class the London Whale case concedes is not catchable. The one not caught is instructive: a spreadsheet dividing by a sum where the author meant an average returns EARNED, because the formula is well-formed and there is no second copy of the intention to compare it to.

This is not a fraud detector. It is an instrument against losing the thread.

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3. Containment, measured

The remaining sections concern collectives of agents whose conclusions rest on one another — the shape of a decentralised autonomous system.

These are separate models, not one experiment. §§3.1–3.3 run 100,000 agents over 1,000,000 links; §§3.4–3.5 run 40,000; §3.6 runs 20,000 with an explicit source layer. Figures are comparable within a subsection and not across them, and no claim below depends on comparing two subsections' absolute numbers.

3.1 Speed is not the problem, and the answer is useless anyway

At 100,000 agents and 1,000,000 dependency links a withdrawal propagates in tens of milliseconds. With one ground per conclusion, a single compromised agent takes down a **median of half the collective** — median here, and worst-case in §3.2, which is not the same statistic and is why §3.2 says so explicitly. "About 59,716 of your conclusions no longer stand" is not something anyone can act on.

3.2 Redundancy contains a random loss and not a chosen one

A conclusion resting on two declared-independent grounds falls only if both do. Sweeping the share of such conclusions gives two different answers depending on where the loss is aimed, and reporting only the first was this work's largest error.

topology	0%	75%	90%	95%
random-local, worst of 25 sampled targets	95,041	51,389	56	4
random-local, **adversary chooses**	99,995	99,984	99,984	99,987
hierarchy, worst of 25 sampled targets	95,081	195	10	8
hierarchy, **adversary chooses**	100,000	100,000	100,000	100,000
scale-free, worst of 25 sampled targets	33,943	142	20	8
scale-free, **adversary chooses**	100,000	100,000	100,000	100,000

Against uniformly sampled targets redundancy works as advertised. Against an adversary choosing the root or the largest hub, **the collective loses everything at every level of redundancy, including 95%.**

There is therefore no threshold against a chosen target. There is one against a random target, and an earlier draft of this note reported the second as though it were the first, because the estimator sampled 25 nodes out of 100,000 and a hierarchy's commander is not among them.

The reason is not a defect in redundancy as an idea. The second ground is drawn from inside the same structure, so it descends from the same root; removing the root removes the alternative with it. That is §3.5 arriving from the other direction — **redundancy that is not independent of the thing being attacked is not redundancy** — and it means the ordinary failure (an agent lost to terrain, a sensor to weather) and the adversarial one need different mechanisms rather than more of the same one.

3.3 Declared redundancy resting on a shared origin

Hold redundancy at 90% and raise the share of "independent" pairs that secretly share an origin. **The collective's own report never moves** — 90% at every row:

secretly shared	random target	top hub	top three
0%	29	1	15
10%	4,102	18	49
25%	29,391	3,793	20,099
100%	99,992	94,180	94,180

A quarter of nominal redundancy being fake destroys containment while the system reports 90%. This is common-cause failure in the reliability engineer's sense (§5a) and is claimed here as a measurement in a warrant setting, not as a discovery. Nothing in the ledger distinguishes the first row from the last, and neither would an attestation layer: both grounds are genuine documents from genuine agents that happen to rest on the same thing.

3.4 Evidence and authority are not interchangeable

A conclusion needs evidence to be *supported* and authority to be *permitted*, so it falls when either fails. Separating the two:

- both dimensions 100% genuinely redundant: losing the commander still costs **C = 1.000** — the entire collective.

An earlier draft reported **C = 0.789** for the case where evidence is redundant and authority is not. A review called that figure invented, this note recorded it as invented, and **both were wrong** — which took a fourth reading to establish and is the reason the whole table is printed here:

configuration	commander lost	mid-level agent lost
evidence redundant, authority not	1.000	**0.1316**
authority redundant, evidence not	1.000	**0.7891**
both redundant	1.000	0.0000

`db/probe_criterion.py` prints **0.7891**. The figure was never invented; it was read off the wrong row — it belongs to *authority* redundant and *evidence* not, the mirror of the case it was attached to. And "the probe prints 1.000 for both configurations" is true only when the lost node is the commander, which is a different column. So the original defect was a mislabelled configuration, the first correction replaced it with a false accusation of fabrication, and this is the third statement of the same three numbers. It is left in full because a corpus that advertises its self-corrections owes the reader the one that was itself wrong.

The dimensions do not average and do not exchange. At an identical minimum redundancy of 0.50, an authority-rich collective loses 0.0003 of itself where an evidence-rich one loses 0.0101 — a factor of 31 — because permission is hierarchically concentrated while evidence is diffuse.

Three outputs, per configuration — not four. **C** (share lost), **r_eff*** (minimal EFFECTIVE redundancy for $C < 1\%$) and **A_crit** (worst single loss). An earlier draft reported r* and q* as two quantities; a later one said they were exactly one, and **that correction was itself too strong**. The argument stands: a conclusion is genuinely redundant with probability $r(1-q)$, so containment should depend on that product alone. The measurement does not support the word *exactly*. On the shipped seed the two crossings land in the same cell of the sweep's own 0.05 grid and **separate by 0.04 once the step is 0.01** — the probe's `1e-9` assert was comparing two numbers quantised to 0.05, so it could not see a gap smaller than its own grid.

Nor is the coarse agreement a property of the grid: it **holds on 2 of 7 seeds tried**, and this note said otherwise for two hours, which is the same error as refuted prediction 10 committed while correcting refuted prediction 9. Both step sizes and all seven seeds are now swept and printed. Measured: **r_eff* ≈ 0.65** across two dimensions, rising to 0.75 when shared models and shared sensor feeds are added — a world with more things to share has more ways to fail.

A_crit is not a function of redundancy at all. It is a property of where authority is concentrated, and is reported beside r* rather than folded into it.

3.5 Authority-root diversity

Varying the number of *independent* authority roots:

configuration	A_crit
1 authority root	**1.000**
2 roots, either suffices	0.117
3 roots, quorum 2-of-3	0.117
3 roots, **shared upstream**	**1.000**

These are closed forms, not measurements: the model has no stochastic content at all — its `rnd` argument is unused — and 0.117 is exactly 4681/40000, the subtree of one first-level commander under a branching factor of 8. Set the branching factor to 20 and the same closed form gives roughly a twentieth (derived, not run). What the table establishes is the structural fact that one root is a single point of failure and two are not; the middle figures are arithmetic about a tree that was chosen.

It stops at two, and only if the two are really two. Three roots hanging off one hidden super-root behave exactly like one, and the report cannot tell the rows apart. A pair and a 2-of-3 quorum lose identically to one loss; the quorum earns its keep against a root that is *captured* rather than lost.

3.6 An incomplete map, and a stale one

When edges are missing from the observed graph the error has a direction, and **which** direction depends on what kind of edge is missing. An earlier draft of this note claimed one direction for incompleteness as such; that was wrong, and both tables are below.

A missing **shared origin** flatters — grounds look more independent than they are:

hidden	C_predicted	C_actual	error
0%	0.089	0.089	+0.000
5%	0.086	0.089	+0.004
50%	0.064	0.089	+0.025

A missing **alternative** frightens — a conclusion the system believes lost is in fact still standing on its other ground:

hidden alternatives	C_predicted	C_actual	error
0%	0.1444	0.1444	+0.0000
50%	0.1734	0.1444	−0.0290
100%	0.2031	0.1444	−0.0587

The note's own §1 supports alternative grounds, so both cases live inside its own model, and the claim that incompleteness is one-directional survived only because one of the two was never run.

The two errors are not equally serious, and that is what survives: an optimistic error lets a system act on warrant it does not hold; a pessimistic one makes it withdraw from warrant it does. The first causes accidents, the second idleness.

The magnitude in the first table is modest — a 28% relative underestimate at full hiding — and the reason matters more than the number: the gap can only be as large as the share of the collective whose fate rides on the *hidden* edges. Five per cent of the wrong edges beats fifty per cent of harmless ones.

Under delay and loss, a collective acting without a gate spends **about two thirds of its actions outside its own containment requirement** — 66% even at zero loss, because delay alone suffices — and nothing reports it at the time.

A gate that acts only while a bound built from observables clears the limit holds violations at zero on three channels and **leaks on two** (270 and 170 violations). The margin used is therefore *not a sound bound*; it is a plausible expression that works in some regimes. A gate right most of the time is not a gate, and the fix is a margin derived from a model of the channel rather than a larger constant. That derivation is not in this work.

3.7 Are these figures, or single observations?

Ten independent seeds, and the answer differs sharply by quantity.

figure	across ten seeds
r_eff* minimal effective redundancy	median 0.725, range 0.6–0.75
redundancy threshold, hierarchy (sampled targets)	median 0.75, range 0.50–0.75
A_crit, one / two / three-on-shared roots	**not a variance result**

The last row is a correction rather than a figure. `probe_roots` is deterministic — its random-number argument is accepted and never used — so running it under ten seeds measured one graph ten times. It printed "constant", which is true and vacuous, and was read as robustness for a day until adversarial review pointed at the function signature.

Ten seeds of one model is still one model. This answers whether a figure is an artefact of a lucky draw; it does not make it general, and it does not apply at all to a model that has no draw in it.

3.8 A graph nobody designed

Every result above was measured on graphs the author generated. Real ones at this scale are ordinary — a Debian installation carries one on disk — and the omission had no defence. The machine's own package database — some two and a half thousand packages and twelve thousand requirement groups on the author's laptop, read 2026-08-17; the program reads whatever host it runs on, and the column below is given in bands for a reason stated after the table:

	synthetic probes	Debian, this machine
edges per node	10.00	**about half that**
declared alternatives	swept 0–95%	**under 3%**
median in-degree	—	1 (top node: over 1,500)
shape	two of three hierarchical / local	heavy-tailed
A_crit	1.000 (one root)	**over five sixths** (`libgcc-s1`)

Why bands and not digits. An earlier draft gave this column to three decimals. Installing a database on the same laptop, hours later and for an unrelated reason, moved every entry in it: the package count rose by 39 and `libgcc-s1`'s share went from 0.868 to 0.870. Neither reading was wrong. The quantity is a property of one host at one moment, and pinning its digits in a document is quoting a ground that expires without notice — this note's own subject, arriving uninvited. The exact figures for any host are printed by `db/probe_real.py`, and that is where a number that moves belongs.

What it confirms. The phenomenon is real and is not an artefact of a generated tree: one package carries most of a working system, and losing it takes that share with it.

What it unsettles, which is more. The synthetic collectives are twice as dense. Two of the three topologies are shapes this graph does not have — and they are the two that carried the threshold in §3.2. Above all, **real declared redundancy here is** under 3% of requirement groups**, while §3.2 located containment somewhere in the seventies. If this graph is at all typical, the region where the threshold was found is a region real systems do not occupy, and the sweep was largely hypothetical.

And one thing does NOT transfer, corrected after review. An earlier version of this paragraph read Debian's `|` as this corpus's `|` and concluded that two fields had reached the same mark and stopped at the same wall. They have not. Debian's `|` is an **ordered preference list over interchangeable providers** — `libcurl3-gnutls |`

`libcurl3-nss | libcurl4` are one library with different backends — and it declares nothing about independence, so it cannot fail to verify one. The notation coincides; the claim does not.

This withdrawal was made in `db/probe_real.py` *on 2026-08-17 and the commit message announcing it said it had been applied "in the probe and in both notes". It had not been applied here; the paragraph above stood for another seven hours, including the words "without qualification". The commit message was false, and it is left recorded because a corpus that logs its refuted predictions should also log the one time its log lied.*

What survives is smaller and still worth having: the share measured above is how often a real system offers ANY choice of provider at all, which bounds from above how much genuine redundancy such a graph could have even if every alternative listed were independent — far below the range §3.3 swept.

3.9 The parameters were carrying the conclusions

Anchoring the sweep on §3.8's real numbers rather than on chosen ones:

density	alternatives	worst sampled	worst chosen	
10.00	90%	0.013	1.000	as swept
10.00	2.6%	0.933	1.000	real redundancy
5.18	90%	0.000	0.977	real density
5.18	**2.6%**	**0.926**	**0.994**	**both real**

At a density and a redundancy a real dependency graph actually has, **there is no containment**: a randomly located loss costs 93% of the collective, because a per-node redundancy that low is indistinguishable from none.

And the redundancy parameter is in the wrong denominator, which review found and which is stated here rather than absorbed. The `2.6%` fed to the sweep is the share of *requirement groups* offering an alternative; the model consumes it per *conclusion node*. Measured on the same package database, the per-node figures are **9.9%** of packages carrying at least one alternative and **0.8%** where every group has one. So this row is anchored on a real number used in the wrong sense — about four times too small in the model's own units, or three times too large against the stricter reading. The conclusion survives the correction, because the sweep is flat here: the *sampled* column reads 0.930 at 0% alternatives and 0.909 at 10%, and the *chosen* column is 0.994 at both. What does not survive is the phrase "anchored on real parameters" without this paragraph beside it.

The sentence above named the wrong column on its first writing — the sampled figures attributed to the chosen one, which is precisely the confusion review had just found in the abstract. It was caught by re-running the probe before committing rather than by reading, and that is the only method that has worked today. Containment appears only above about 75% redundancy, and real systems sit at the other end of that axis.

Density turned out not to be the dial an earlier draft of the probe claimed: the collective is uncontained at every density tried, from 2 edges per node (0.719) upward, saturating by about 3. That was a third conclusion written before its table was read, and it is recorded in the source with the others.

And the chosen-target column barely moves, with a floor of 0.791. Across every row of every table in §§3.2 and 3.9, an adversary aiming at the root takes at least that share of the collective, and at every density above 3 edges per node at least 0.94. The sensitivity is to whether the attacker chooses, not to density or redundancy. It is the most robust result in this note, and it is a negative one.

This paragraph said the column "does not move anywhere ... at any density and any redundancy" until 2026-08-17, while the sweep it summarises printed 0.791 at density 2. The abstract was corrected first and this sentence was not

— the third time in one day that a withdrawal was applied where it was found and left standing where it was not, which is why `inventory/withdrawn_claims.py` now carries a signature for it.

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4. Five things a system must know, and what these runs measure about them

The measurements above separate into a ladder — arrived at in correspondence rather than derived from any single run, and **three of its five rungs have a literature that predates it by decades** (§5a): authority separated from evidential support, and currentness as a bounded recency requirement, are both formalised in the authentication and authorization logics of 1989–1999. It is stated here because it organises what the runs found, not because it is new:

1. **identity** — who or what signed a ground;
2. **provenance** — what it descended from;
3. **independence** — whether two grounds may count as two;
4. **bounded completeness** — whether the map is finished *enough*, for this scope and this decision;
5. **currentness** — whether that judgement is still true now.

Attestation supplies (1) and provenance systems (2). Of the remaining three, **currentness (5) has a literature this note had not read** — Burrows, Abadi and Needham formalise freshness in 1989 and Stubblebine [9] is an entire paper on bounding a credential's staleness (§5a) — and **bounded completeness (4) is the regulatory category of completeness uncertainty**, NUREG-1855 §2.3.3, in the same corpus this note already cites at [15, 16]. An earlier sentence here read "nothing found supplies (3), (4) or (5)", contradicting this section's own opening line thirteen lines above; the contradiction was prescribed for repair by a review two rounds ago and applied everywhere except here. What the runs establish is what each rung is worth, not that anyone lacks it: q* says containment survives about 35% hidden correlation and no more; the blindspot **tables** — two of them, and that is the point — say the error from incompleteness has a direction fixed by the KIND of missing edge, a hidden shared origin flattering and a hidden alternative frightening (§3.6); the gate says a warrant is usable only while its staleness can be bounded.

Stated as control rather than observation, and narrowly:

> A system may continue to **treat** its current warrant as satisfying the stated containment criterion only while it can bound that warrant's possible staleness within that criterion.

Not *may act*, and not *may rely* either: both are permissions, and nothing here establishes one. Whether reliance or action is allowed in that state is decided by an external authority layer. This work measures the antecedent and does not reach the consequent — the same boundary it draws everywhere else, which took three drafts to stop slipping in the one sentence meant to summarise it.

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5. Predictions that failed

Reported because a corpus that hides them measures nothing. Each is pinned by an assertion in the source so it cannot be quietly tuned away.

1. *"The minimum over dimensions governs containment."* Refuted by its own table: at an identical minimum, a factor of 31 between configurations.

2. *"A second authority root makes the worst loss negligible."* It is 0.117. The concentration moved from the root to a mid-level agent rather than dissolving.
3. *"Lag zero is not safe."* It is. At zero lag the blind window is zero.
4. *"Hiding 5% of dependency edges takes a predicted 1% to an actual 40%."* The direction held; the magnitude did not — 28% relative.
5. *The runtime gate.* Zero violations on three channels, leaks on two.
6. *"An autonomous collective may act on a warrant only while…"* A normative claim this work does not establish. Corrected twice: *rely* is also a permission, and the descriptive form is *treat as satisfying*.

Two methodological errors are also recorded in the source: a first blindspot model whose shared structure was too redundant for hiding to show, and a median that concealed the hub risk it was measuring.

Six more were found by an adversarial review of this note rather than by its author, and they were the expensive ones. They are listed here because a note that reports only the failures its author noticed is reporting a selection.

7. *"The table in §3.2 is worst-case."* It was the worst of 25 uniformly sampled targets out of 100,000. The true worst — a chosen root — is 100,000 at every level of redundancy, and the note had claimed a correction to exactly this error while continuing to make it.
8. *"Redundancy means two grounds."* The code appended a parent and switched the node's entire input list from OR to AND. The flagged nodes had a median of eleven inputs and **none** had exactly two, so every figure quoted from that operation was about something else. The semantics is now what the prose always said, and §3.2's numbers changed accordingly.
9. *"r* and q* are two quantities."* Read from two ends they nearly coincide, because effective redundancy is r(1-q) — but the follow-up claim that `1 - q* = r*` holds *exactly* is refuted too (§3.4): the two crossings coincide only at the sweep's own 0.05 step.
10. *"A_crit is constant across ten seeds."* The model is deterministic and its random argument unused; ten seeds measured one graph ten times. 0.117 is 4681/40000, a closed form in the branching factor.
11. *"C = 0.789 was invented; no program produces it."* **This entry was itself refuted, 2026-08-17.** `probe_criterion` prints 0.7891 — the figure is real and was attached to the mirror-image configuration (§3.4). A list of refuted predictions that contains a refuted refutation is the honest state of it.
12. *"The error from incompleteness runs one way only."* It runs one way per KIND of edge. A missing shared origin flatters; a missing alternative — which §1 explicitly supports — makes the prediction too large. The claim survived because only one of the two was ever run.

Adversarial review raised 48 findings of which 32 survived a refutation pass; the full list is in the repository beside this note. The reader should assume the same density of error in whatever has not yet been reviewed.

5a. Neighbours, and what was and was not searched

Common-cause failure analysis is the nearest neighbour of §3.3, and was missed in the first draft of this section — the more embarrassing omission of the two, because it is closer than the one that was cited. Reliability engineering has studied since the 1970s exactly the phenomenon §3.3 reports: redundancy defeated by a dependency shared between the redundant parts. The β-factor is the fraction of failures arising from a single common cause. This paragraph called it "the same quantity this note calls hidden correlation `q`" — **and the paragraph below withdraws exactly that equation**, on the ground that the two are not commensurable. Both stood in this section for a day. The withdrawal is the correct one and this sentence is corrected to match it: β and `q` occupy the same *role* in their respective models and are not the same quantity. IEC 61508 standardises its estimation (a 37-question checklist over eight classes of defensive measure, among them *diversity/redundancy* and *separation/segregation*), and fault-tree analysis, common-mode analysis, NUREG methods in nuclear engineering and NASA's work on ultra-reliability all address it.

Two consequences, both against this note. First, §3.3 reports a named, standardised phenomenon and claims no discovery. Second, an earlier draft compared β directly with `q` and called this work's tolerance "an order of magnitude looser" than engineering practice. That comparison is withdrawn: the quantities are not commensurable. β is a conditional fraction of component *failure rates* attributable to a common cause; `q` here is a structural share of *declared redundancy that is fake*. Equating them needs an argument this note does not have. Also, β is not the state of the art — it was superseded precisely because it cannot handle k-of-n redundancy, by the Multiple Greek Letter, Binomial Failure Rate and α-factor models [2].

A sentence here used to begin "What CCF analysis does not do, so far as I can find…". **It is cut.** Four sentences of that form were destroyed in one day — three by installing one package, one by reading a function list — and the common factor is that "so far as I can find" cannot be paid for by finding. What the difference of subject is, without any claim about who lacks what: CCF quantifies the *probability* that components fail together; the measurements here ask what a conclusion is still *warranted* by afterwards. Whether that difference is interesting, and whether it is already someone's subject, is for a reader who knows that literature.

Distributed authorization logic already separates authority from evidential support, which was §5a's first novelty bullet until adversarial review removed it. Abadi, Burrows, Lampson and Plotkin [3] give a modal `says` and a `speaks-for` delegation relation; Lampson, Abadi, Burrows and Wobber [4] separate the authentication chain from the authorization chain. SPKI/SDSI [5] has authorization certificates, delegation-chain reduction and **k-of-n threshold subjects** — which is §3.5's "quorum 2-of-3", standardised in 1999. Decentralized trust management [6] and the RT framework [7] compute credential chains, which is the §3.5 computation. The operational instance of §3.5's last row is also familiar to that community: nominally distinct certificate authorities cross-signing to a shared root.

And "currentness" is not new either. Burrows, Abadi and Needham [8] formalise in 1989 both the jurisdiction postulate (`P controls X` — authority, kept apart from evidential belief) and nonce-verification (freshness), i.e. rungs (1) and (5) of §4 together. Stubblebine [9] is an entire paper on bounding the staleness of a credential as an explicit recency requirement, and certificate revocation has argued about freshness windows ever since [10].

So of the five rungs in §4, three have a literature this note had not read when it named them. What the runs measure is the interaction — but the ladder is a re-derivation, and §4 says so now.

Truth-maintenance systems are the nearest neighbour of §1. Doyle's TMS (1979) and de Kleer's ATMS (1986) already do the central operation here: record which conclusions rest on which assumptions, and recompute what stands when an assumption is retracted. The cascade of §1 is that operation, and this work claims no priority over it.

Nor over **belief revision** (AGM and successors), which studies what a rational agent should give up when a new fact contradicts an old one; over **provenance semirings**, which annotate query results with the tuples that produced them; over **argumentation frameworks**, which compute which arguments survive given attacks between them; or over lineage systems in databases and workflow engines, which track derivation for exactly this purpose.

The next sentence used to read "what is not standard in those, so far as I can find, is the combination…". **It has been narrowed, because the claim rode an atom no search can pay for.** "Not standard" needs the search to have been systematic; this one was thirty minutes with LLM assistance, and §5a says so two paragraphs down. Twice today a bullet of exactly that form was destroyed within the hour by installing one package. So what follows is stated as what it is — **places I looked and did not find, which is weak evidence and is offered as weak evidence**:

- ~~an **authority dimension kept separate from evidence**~~ — **WITHDRAWN 2026-08-17.** It is standard: ProvSQL ships `sr_minmax`, and its own documentation demonstrates it computing the clearance level required to have inferred a derived fact. TMS justifications indeed do not distinguish permission from support; provenance semirings over an ordered carrier do;
- ~~**declared independence that the machine refuses to assume**, reported as a bracket~~ — **WITHDRAWN 2026-08-17.** Tuple-independence is a documented *default* in ProvSQL, not an assumption of the formalism, and one statement yields the dependent reading. Beskales et al. (PVLDB 2(1), 2009) return min/max counts and confidence intervals precisely where record identity is unresolved — this bracket, this problem, seventeen years earlier;
- **hidden common origin** with a measured tolerance (§3.3, q*). I did not find it priced this way in the databases literature; I did NOT search the reliability literature for it, and [15, 16] are exactly where it would be — the β-factor prices correlated failure among components known to be distinct, which is adjacent and may be nearer than adjacent;
- **incompleteness and staleness of the dependency map itself** as measured quantities (§3.6). Not found in the sources read; not searched for in observer design or networked control, where "the state estimate is partial and late" is the standing subject and the nearest thing to a refutation of this bullet would live;
- and **runtime containment under consequential action**, which is a control problem rather than a maintenance one. Naming it as control is precisely an admission that a literature exists which this work has not read.

Two of those five were struck on 2026-08-17, after the free implementation of provenance semirings was installed and asked the same questions rather than only cited (`paper/PROVSQL-REVIEW-FINDINGS.md`). The three that survive have been rewritten from claims into **directions a reader should check**, because that is all the search behind them supports.

And the honest reading of the whole section is that it should not be read as a boundary. Its history: five items, of which two died within an hour of someone installing the neighbouring tool, and three now carry a note saying which literature was not searched. A section whose claims have that mortality rate is evidence about the search, not about the field. It is kept because naming one's neighbours is owed to a reader, and it is stripped of every assertion about what those neighbours lack.

Stated plainly: the mechanism is old. What this note contains is a set of measurements of that mechanism under evidence, authority, hidden common dependencies, an incomplete map and a changing world — and the measurements are mostly negative. A reviewer who reads §1 and thinks "this is an ATMS" is reading it correctly.

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6. Limits

These are simulations on synthetic graphs, and §3.8 measures how unlike a real one they are — twice the density, a shape two of three generators do not share, and a redundancy sweep across a region where real systems sit at under three per cent. The existence of a threshold may be robust; its location is a property of the model, and the model is not close to the one real graph tested.

Citations are honoured, never discovered. A dependence nobody recorded is invisible, and the error it causes runs in a direction set by the kind of edge that is missing: an unrecorded **shared origin** makes a measured blast radius too small, an unrecorded **alternative ground** makes it too large (§3.6, where the second table predicts

0.2031 against an actual 0.1444). The ledger cannot tell which kind it is missing, so neither reading may be assumed.

An earlier version of this paragraph said the radius "can be understated and never overstated". That is the claim §3.6 and refuted prediction 12 withdraw, and it stood here for a day after the withdrawal — in the section a reader consults for the honest boundary, which is the worst place in the document for it to have survived.

Independence between external grounds cannot be verified. Two photocopies of one invoice buy the same immunity as two documents. Where both grounds are claims inside the ledger, the shared ancestor is computed and named; between external papers it is not.

Against an adversary controlling the input the instrument is inert, not merely weak. It grades what it is told.

One failure class is absent by construction: a compromised agent that continues to sign and to lie. This work models *loss*, not malice.

The Lean result and the containment results are two contributions, not one. The calculus is machine-checked; the containment measurements are simulation. Nothing here proves a containment property formally.

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7. Reproduction

Public repository, one command each, no dependencies beyond the standard library:

```
python3 db/probe_ledger.py        python3 db/probe_swarm.py
python3 db/probe_assertions.py    python3 db/probe_topology.py
python3 db/probe_lattice.py       python3 db/probe_containment.py
python3 db/probe_override.py      python3 db/probe_criterion.py
python3 db/probe_failures.py      python3 db/probe_classes.py
python3 db/probe_system_errors.py python3 db/probe_roots.py
python3 db/probe_blindspot.py     python3 db/probe_currentness.py
python3 db/probe_gate.py          python3 run_all.py
```

`run_all.py` runs 121 stands and the Lean corpus, and asserts the zero-axiom line.

References

Assembled after adversarial review pointed out that the first draft carried five novelty claims and no bibliography — three named authors in twenty-one thousand characters. A priority claim with no citation is not a claim, it is an assertion, and "so far as I can find" is unauditable unless the search is described. The search behind §5a was conducted with LLM assistance over the open web and the author's own reading; it is **not** a systematic review, and readers should treat the novelty bullets as an invitation to correct rather than as a survey result.

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Citations 1–17 were checked for author, venue and year; page ranges for 3, 4, 7, 9, 10 and the annex structure of 15 were not independently verified against the printed sources and should be confirmed before this note is cited in turn.

Acknowledgement

The instrument and this note were built with Claude (Opus 5) as architect and implementer, under Variant A, with Vitaly Reznik as human curator. Several of the failed predictions in §5 were the model's, and several of the corrections that found them came from correspondence with colleagues whose contribution will be acknowledged in the joint work that follows.